

Shyamoli Textile Engineering College
Department of Computer Science and Engineering
CSE-4237: Digital Image Processing

Lec-11: Color Image Processing

Digital Image Processing by Rafael C. Gonzalez and Richard Eugene Woods (Chapter-6)

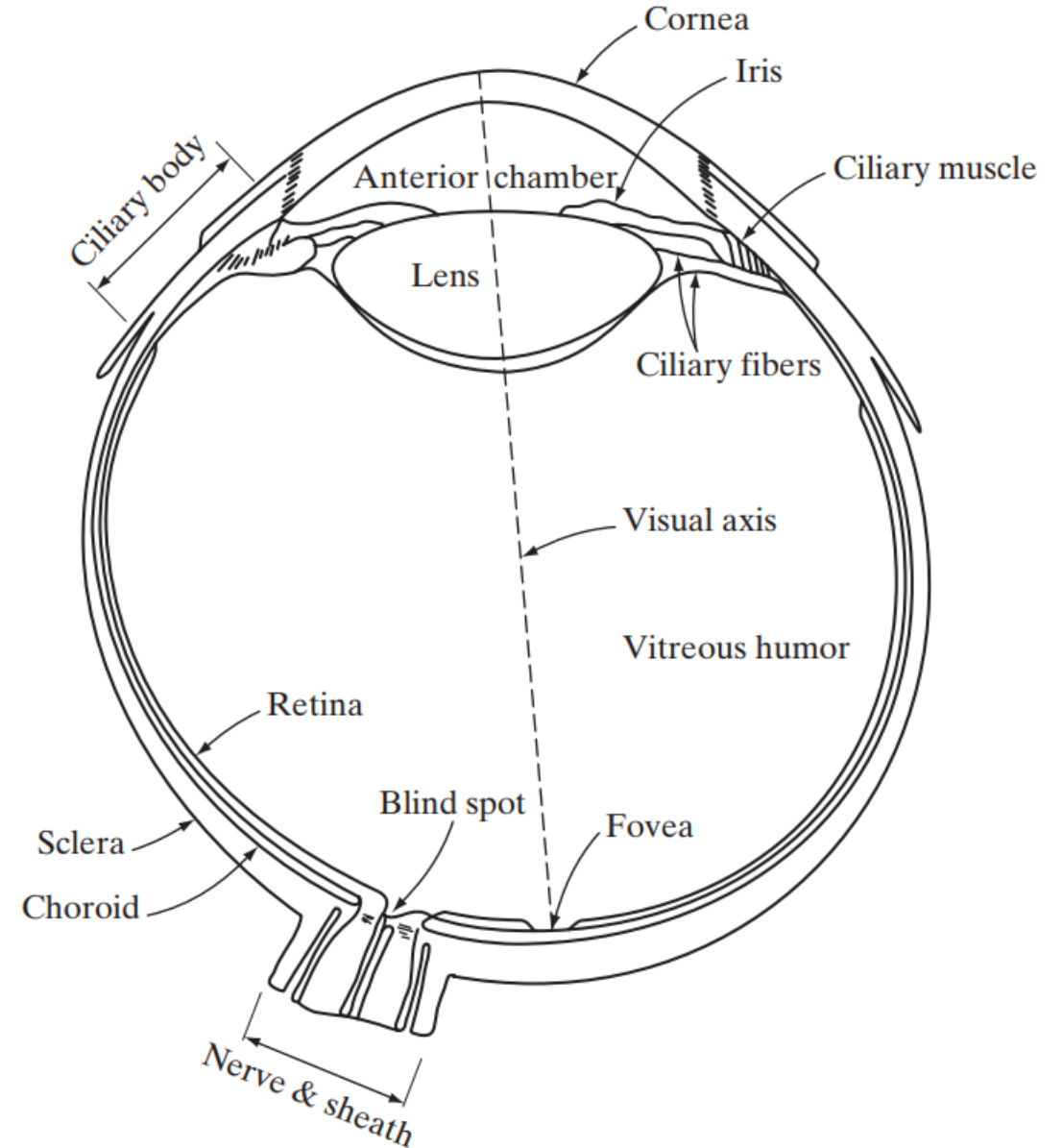
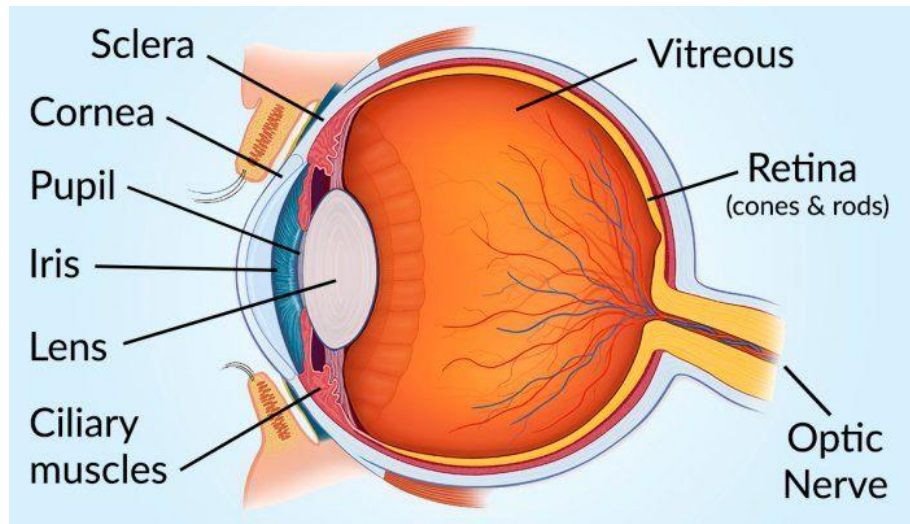
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1. Structure of the Human Eye

The eye is nearly a sphere, with an average diameter of approximately 20 mm.

Three membranes enclose the eye:

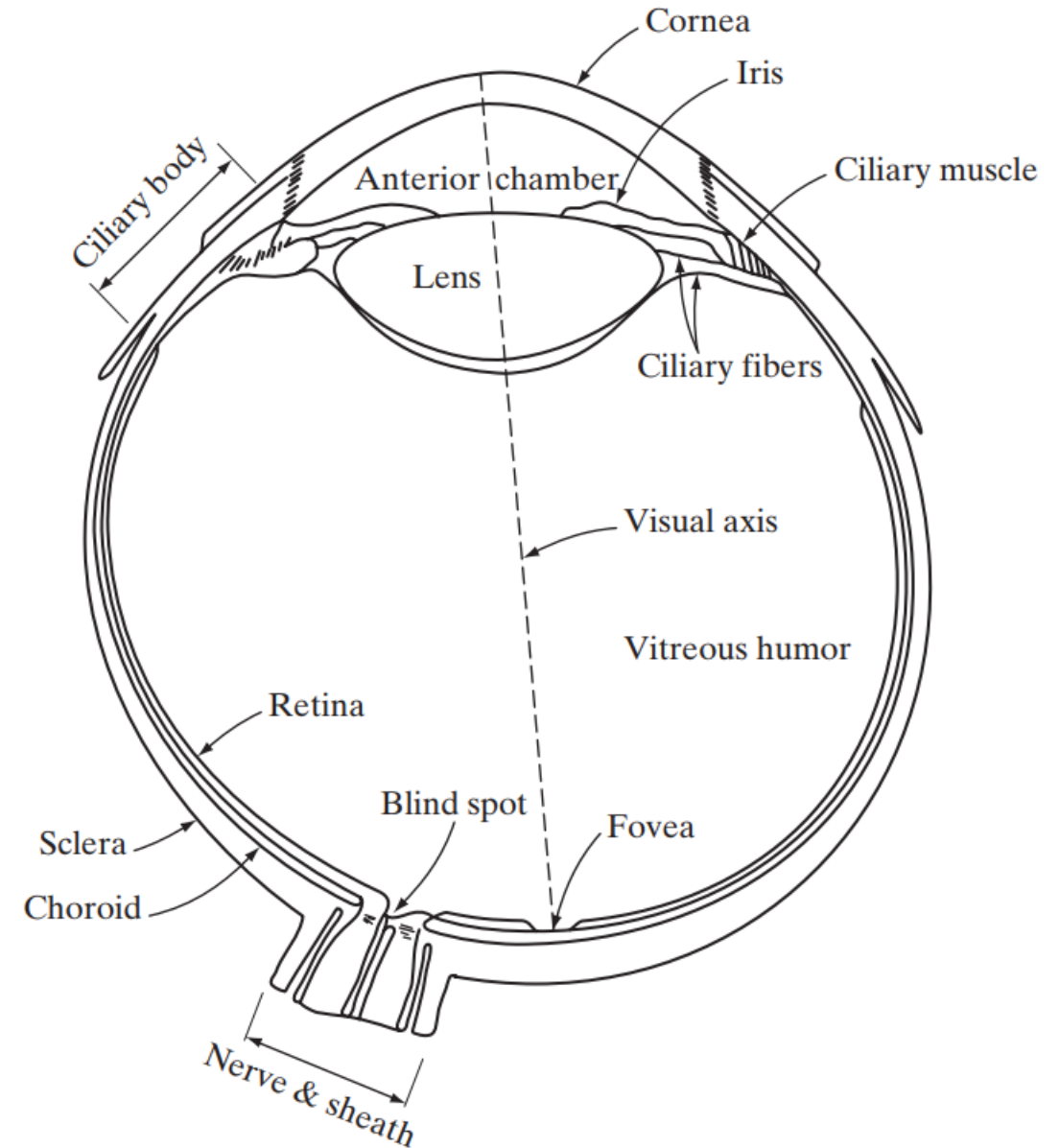
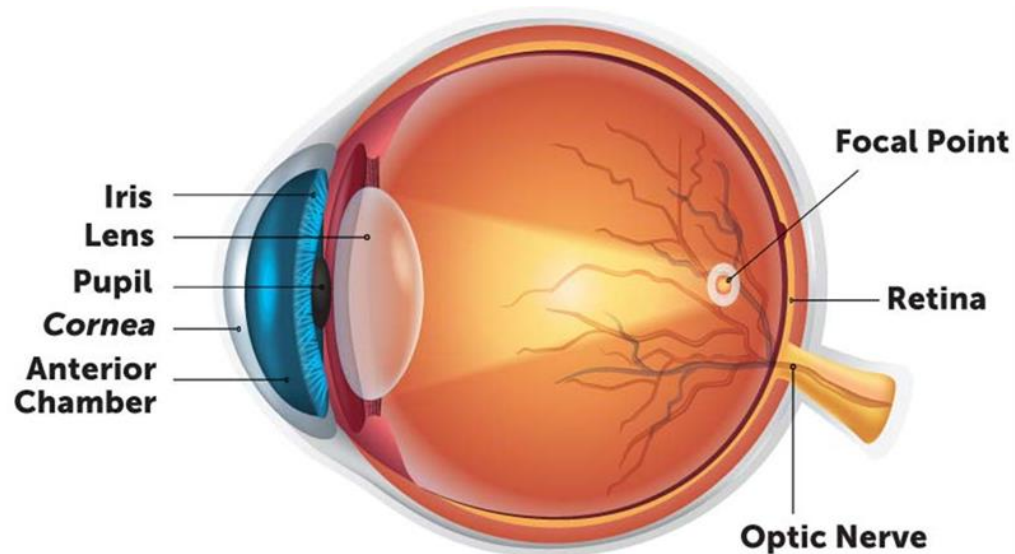
- ❑ the cornea and sclera outer cover;
- ❑ the choroid; and
- ❑ the retina.



1.1 Cornea and Sclera

The cornea is a tough, transparent tissue that covers the anterior surface of the eye.

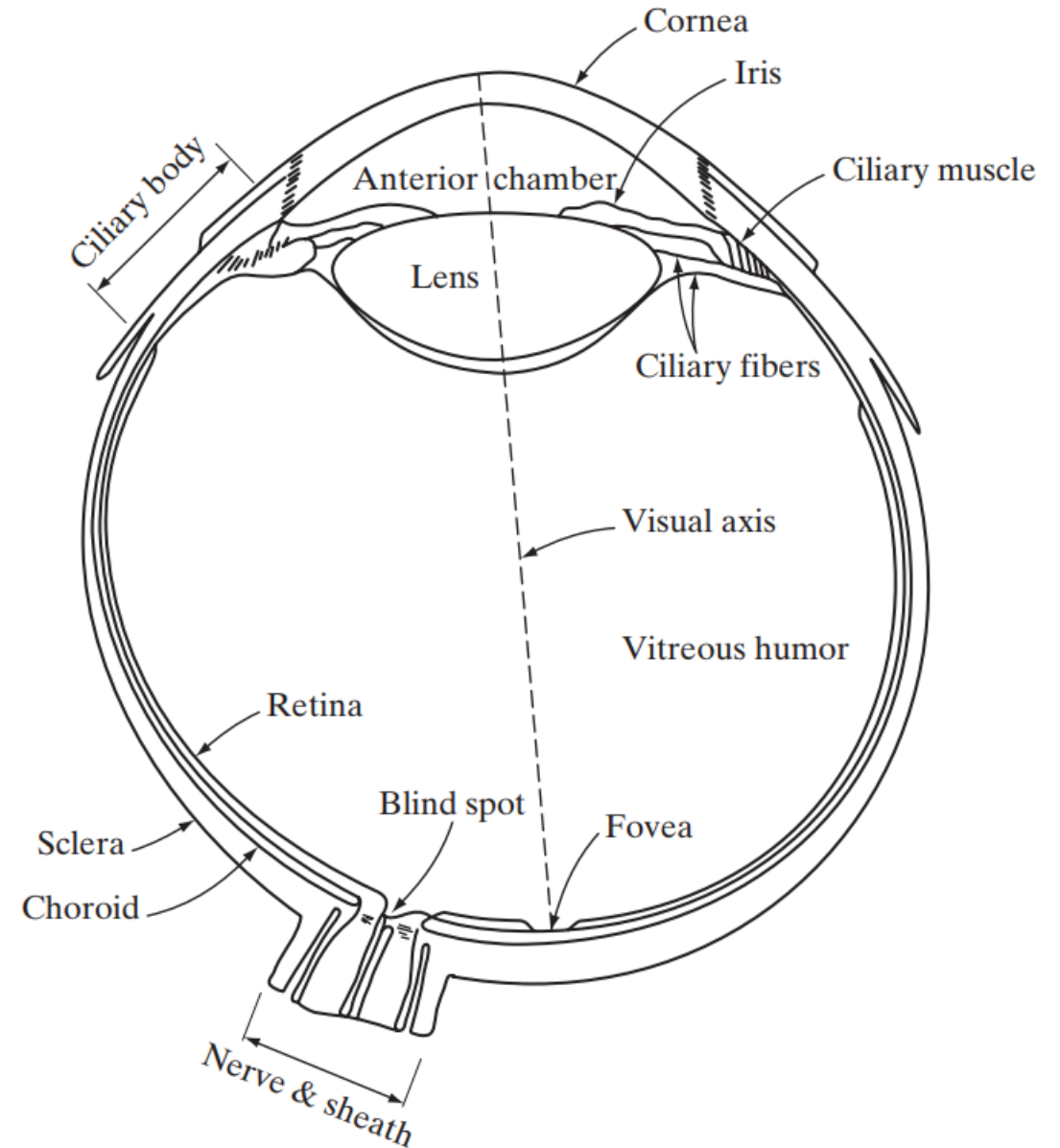
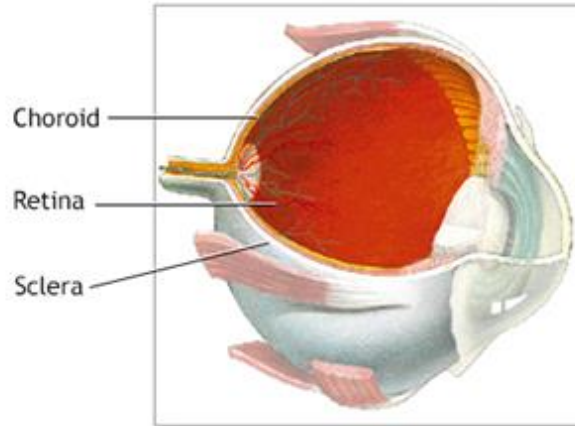
Continuous with the cornea, the sclera is an opaque membrane that encloses the remainder of the optic globe.



1.2 Choroid

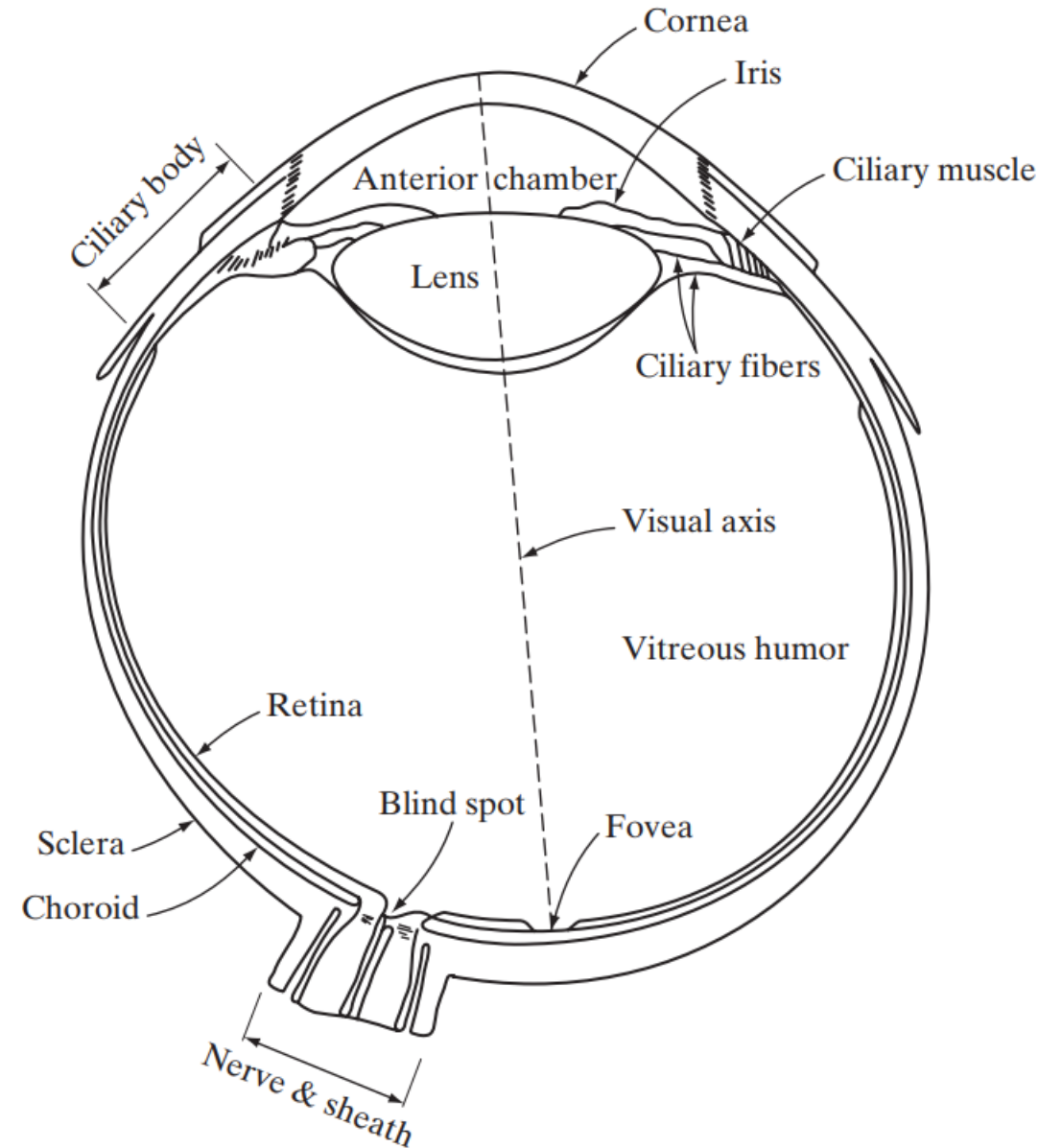
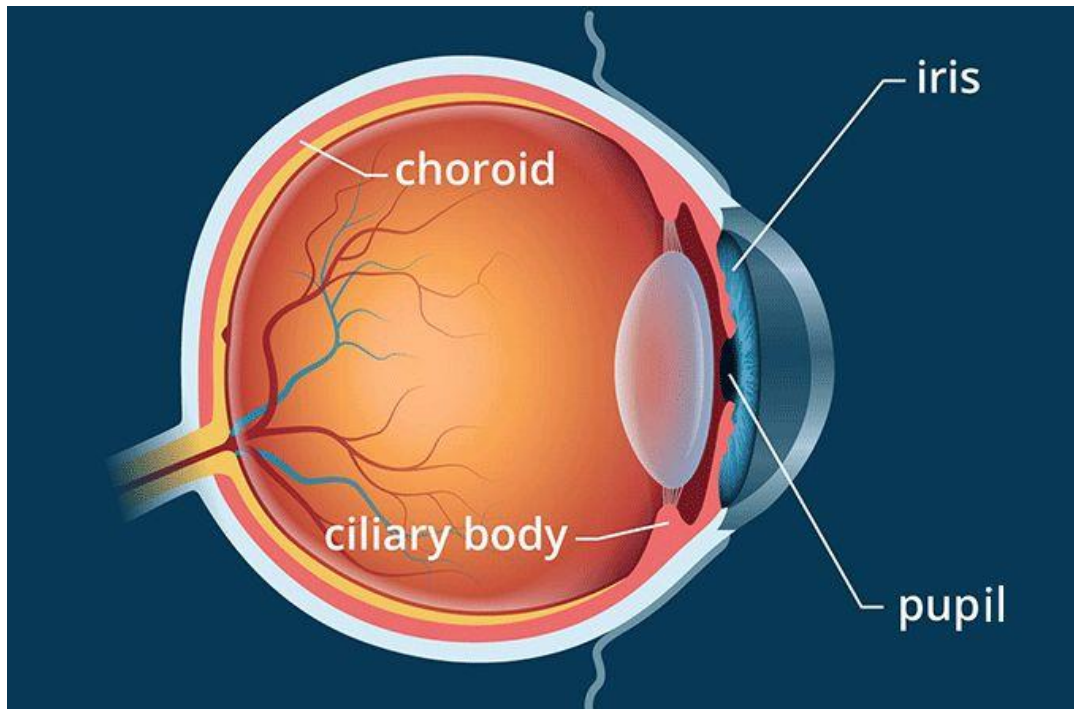
The choroid lies directly below the sclera. This membrane contains a network of blood vessels that serve as the major source of nutrition to the eye.

The choroid coat is heavily pigmented and hence helps to reduce the amount of extraneous light entering the eye and the backscatter within the optic globe.



At its anterior extreme, the choroid is divided into the ciliary body and the iris.

The Iris contracts or expands to control the amount of light that enters the eye.

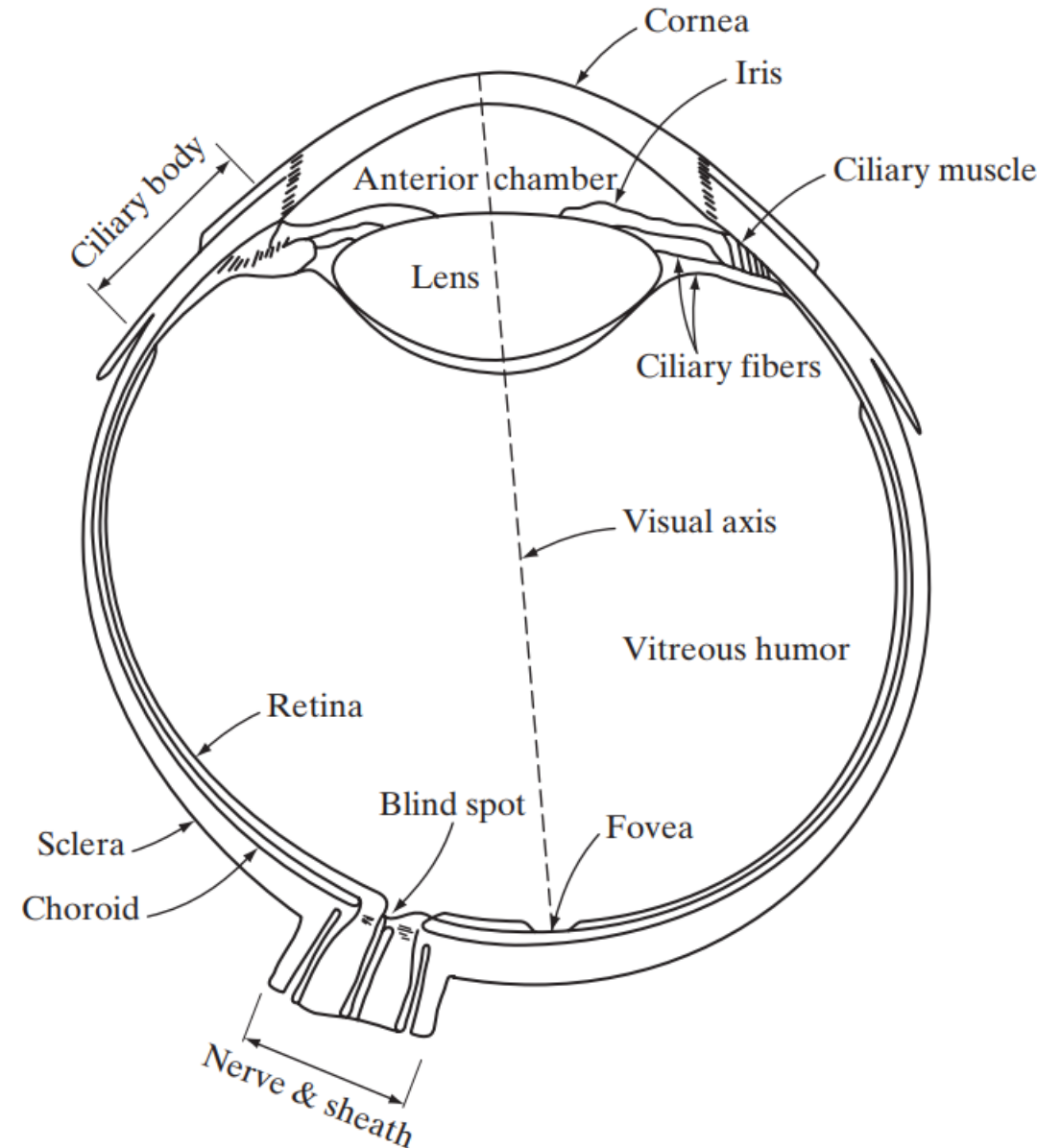


1.3 Lens

The lens is made up of concentric layers of fibrous cells and is suspended by fibers that attach to the ciliary body.

It contains 60 to 70% water, about 6% fat, and more protein than any other tissue in the eye.

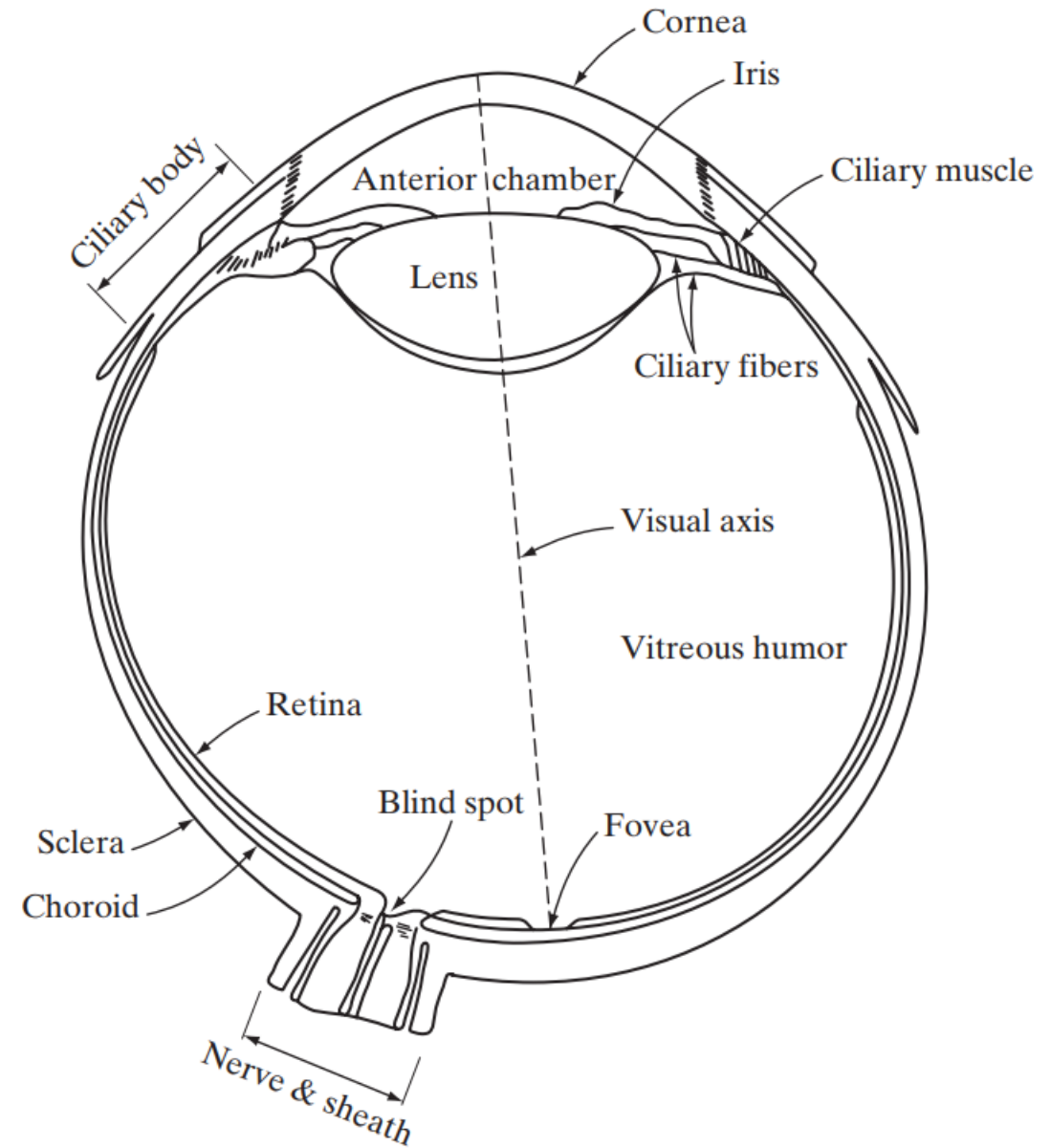
The lens is colored by a slightly yellow pigmentation that increases with age.



In extreme cases, excessive clouding of the lens, caused by the affliction commonly referred to as cataracts, can lead to poor color discrimination and loss of clear vision.

The lens absorbs approximately 8% of the visible light spectrum, with relatively higher absorption at shorter wavelengths.

Both infrared and ultraviolet light are absorbed appreciably by proteins within the lens structure and, in excessive amounts, can damage the eye.

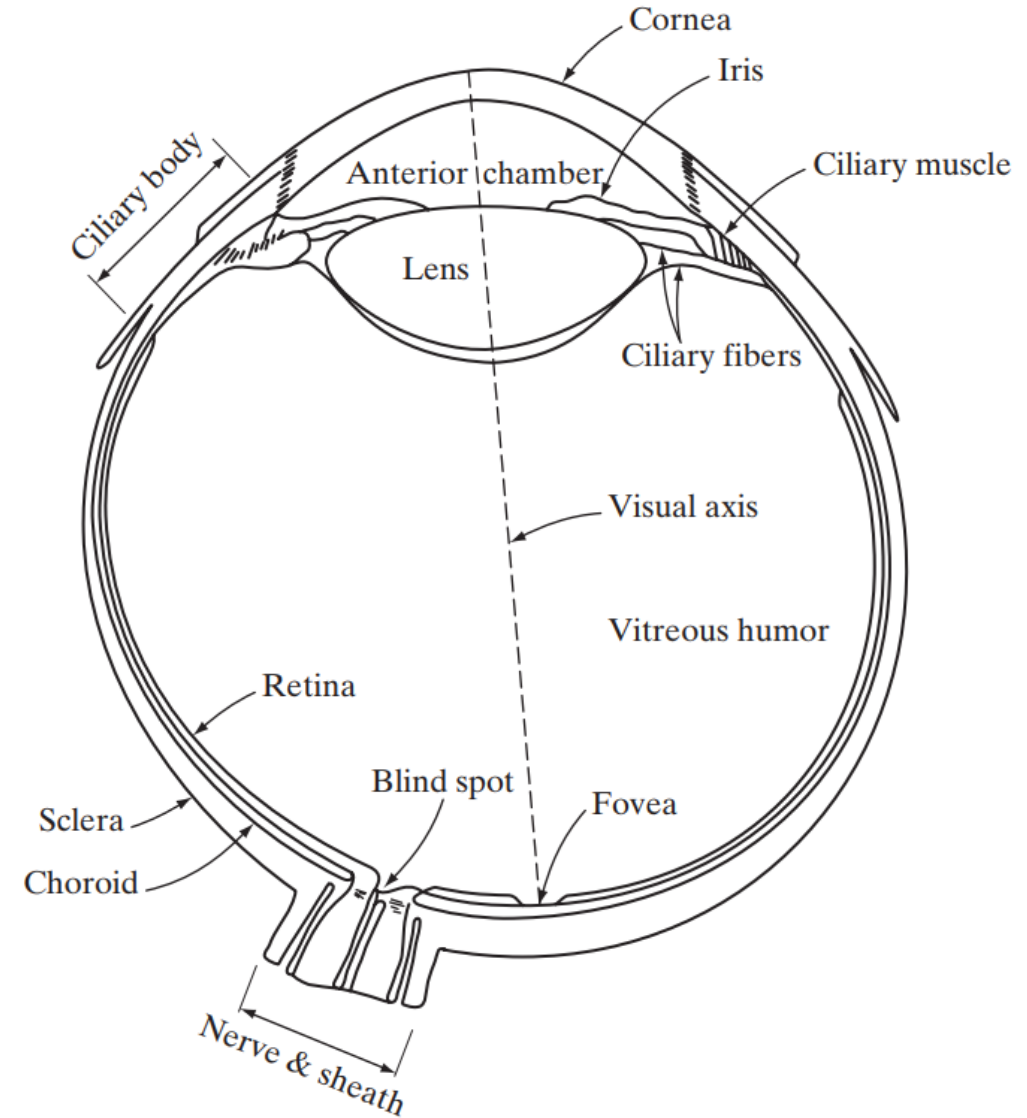


1.4 Retina

The innermost membrane of the eye is the retina, which lines the inside of the wall's entire posterior portion. When the eye is properly focused, light from an object outside the eye is imaged on the retina.

Pattern vision is afforded by the distribution of discrete light receptors over the surface of the retina.

There are two classes of receptors: cones and rods.



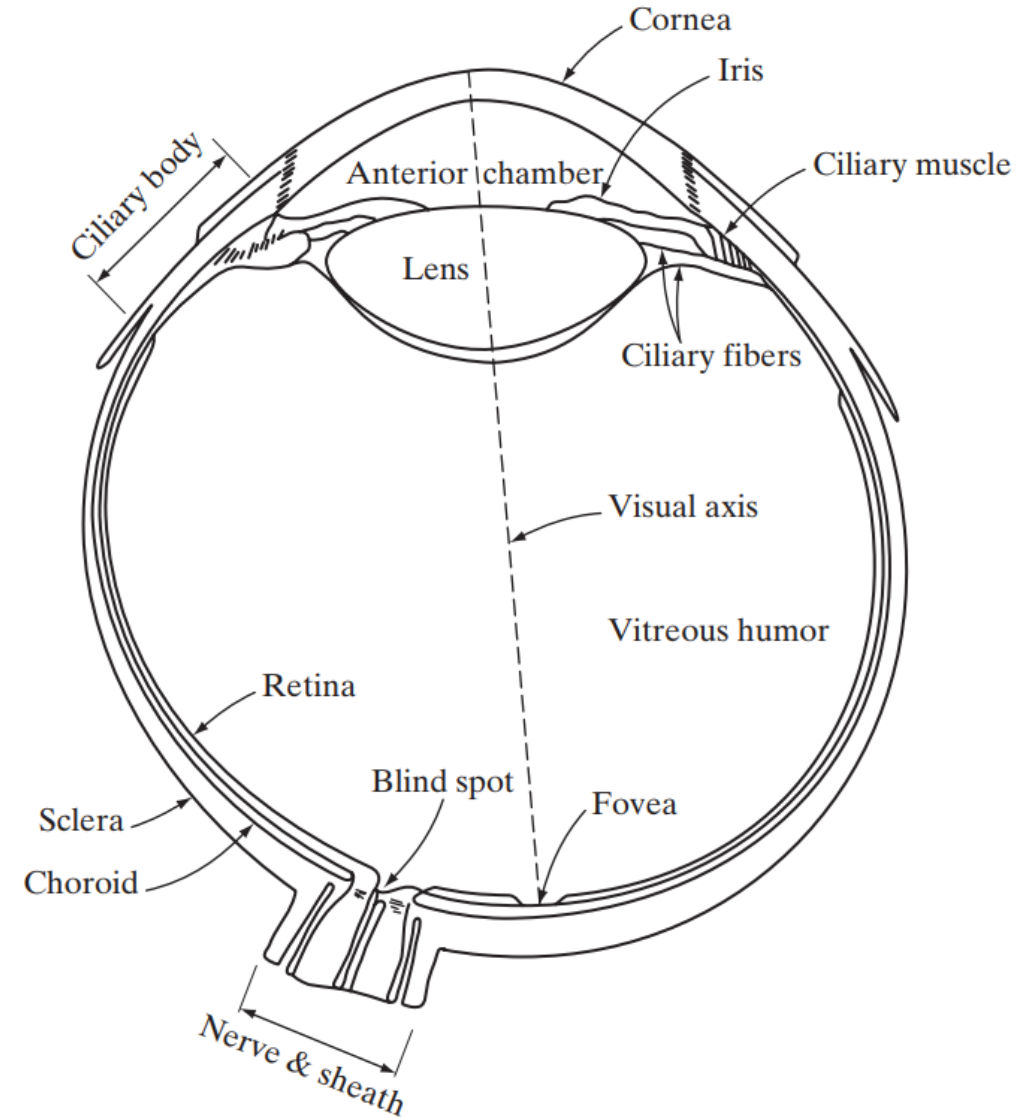
Cones

The cones in each eye number between 6 and 7 million. They are located primarily in the central portion of the retina, called the fovea, and are highly sensitive to color.

Humans can resolve fine details with these cones largely because each one is connected to its own nerve end.

Muscles controlling the eye rotate the eyeball until the image of an object of interest falls on the fovea.

Cone vision is called photopic or bright-light vision.



Rods

The number of rods is much larger: Some 75 to 150 million are distributed over the retinal surface.

The larger area of distribution and the fact that several rods are connected to a single nerve end reduce the amount of detail discernible by these receptors.

Rods serve to give a general, overall picture of the field of view. They are not involved in color vision and are sensitive to low levels of illumination.

Rod vision is called scotopic or dim-light vision

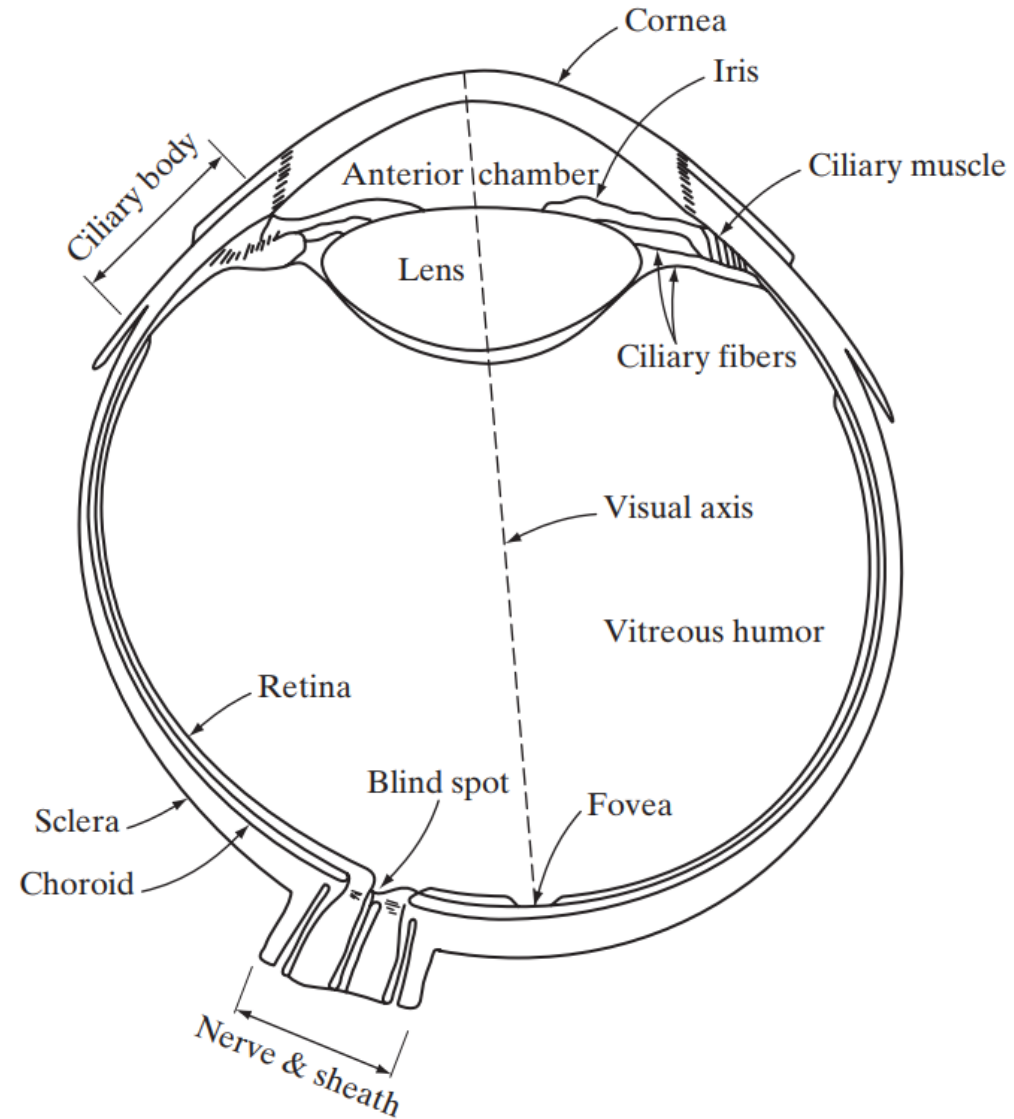
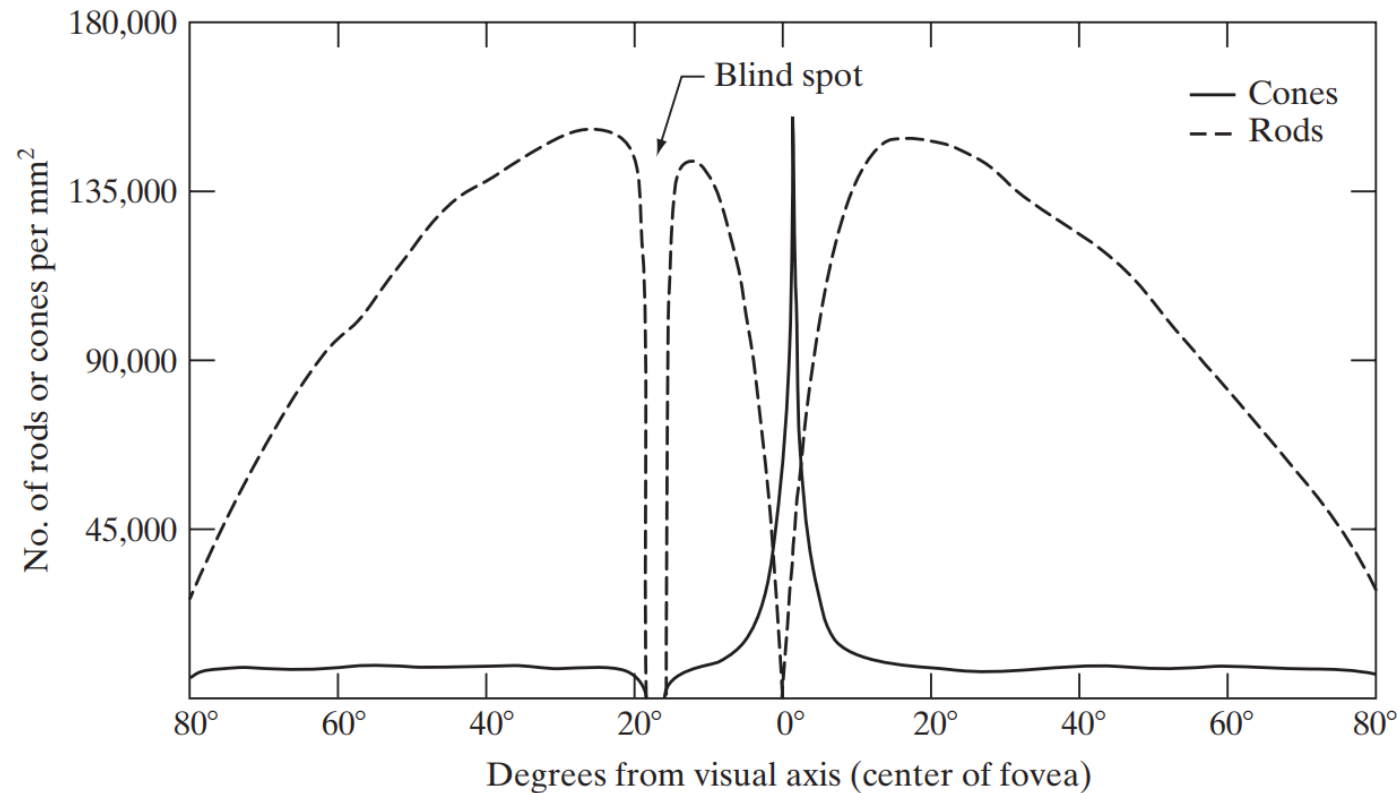


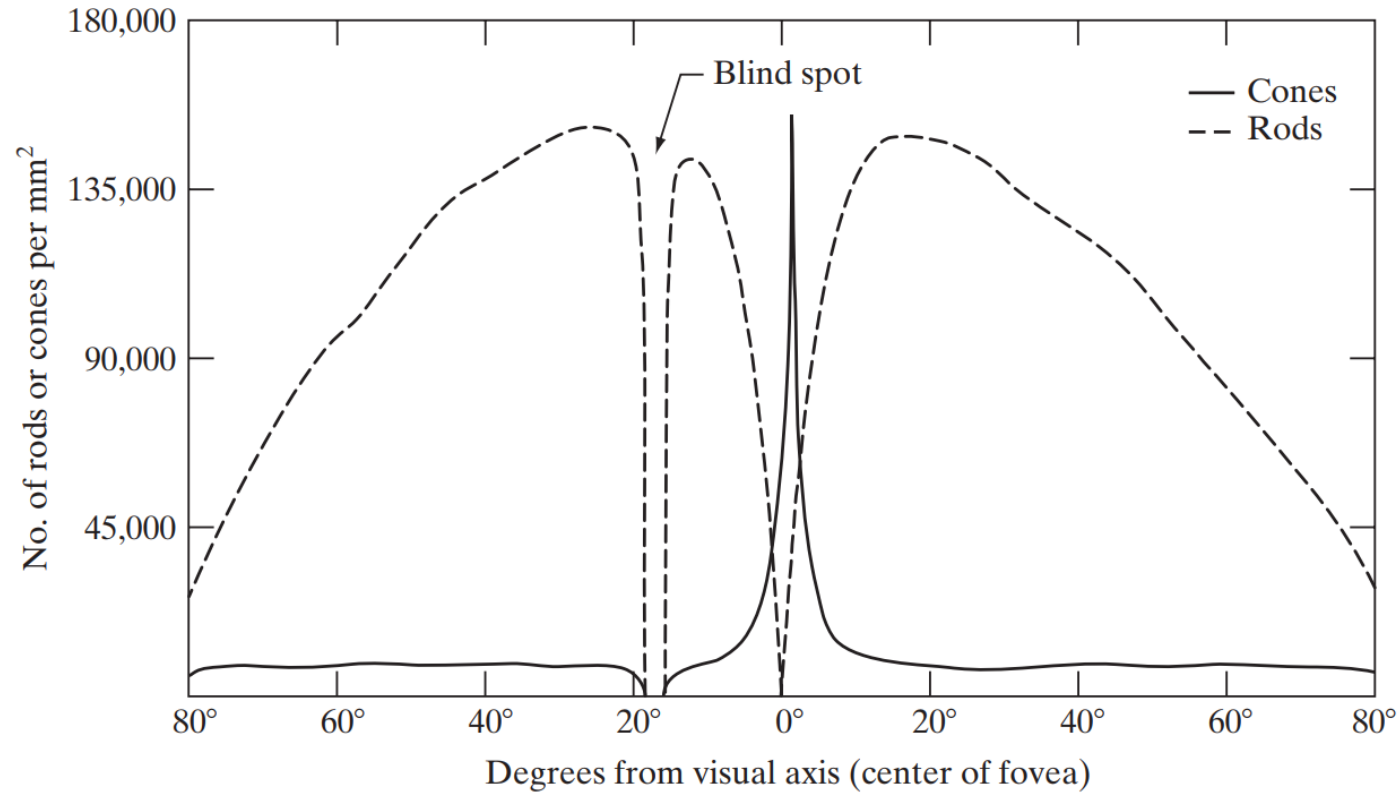
Figure shows the density of rods and cones for a cross section of the right eye passing through the region of emergence of the optic nerve from the eye.



Distribution of rods and cones in the retina.

The absence of receptors in this area results in the so-called blind spot.

Except for this region, the distribution of receptors is radially symmetric about the fovea. Receptor density is measured in degrees from the fovea (that is, in degrees off axis, as measured by the angle formed by the visual axis and a line passing through the center of the lens and intersecting the retina).



Distribution of rods and cones in the retina.

2. Introduction to Color Image Processing

The use of color in image processing is motivated by two principal factors.

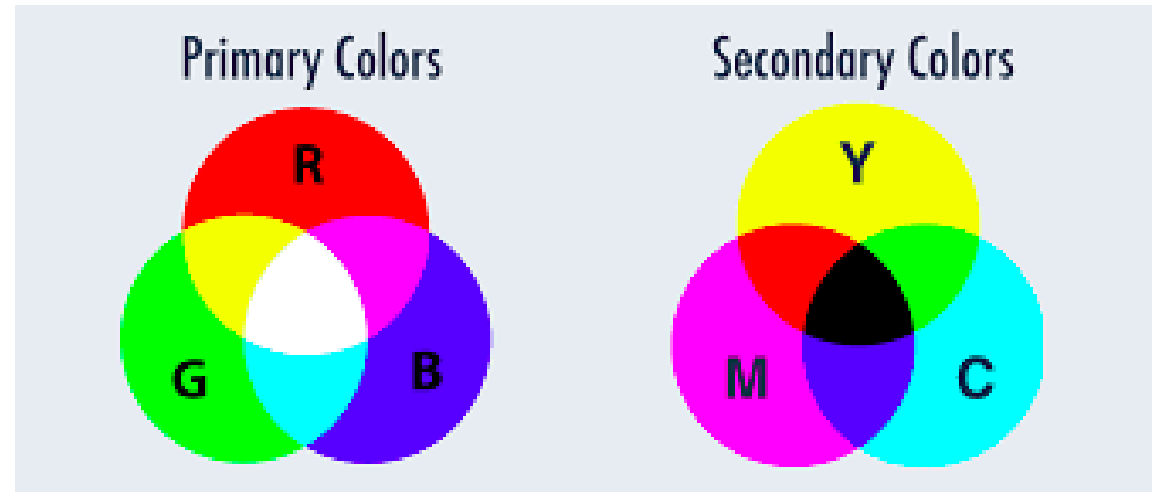
- ❑ First, color is a powerful descriptor that often simplifies object identification and extraction from a scene.
- ❑ Second, humans can discern thousands of color shades and intensities, compared to about only two dozen shades of gray. This second factor is particularly important in manual (i.e., when performed by humans) image analysis.

Color image processing is divided into two major areas: full-color and pseudocolor processing.

2.1 Full-color processing

Full-color processing involves working with images that are captured or displayed in full color, typically using RGB (Red, Green, Blue) color spaces or other color models like CMYK (Cyan, Magenta, Yellow, Black).

This type of processing is used in applications where the true color representation of the image is important, such as in digital photography, video processing, and computer vision.



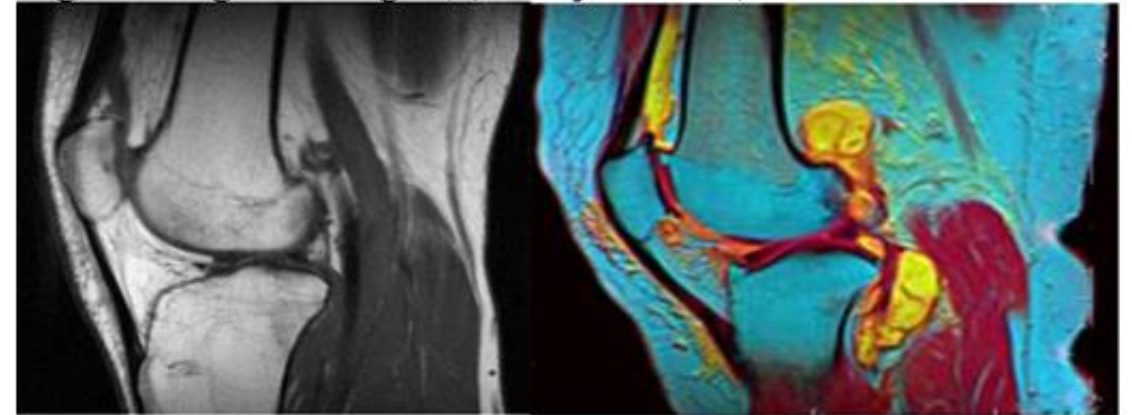
2.2 Pseudocolor processing

Pseudocolor processing involves assigning colors to grayscale images or data to make certain features more visible or to represent data that is not originally in color form.

This technique is often used in medical imaging (like MRI or CT scans), satellite imagery, and scientific visualization where the goal is to highlight specific details or patterns.

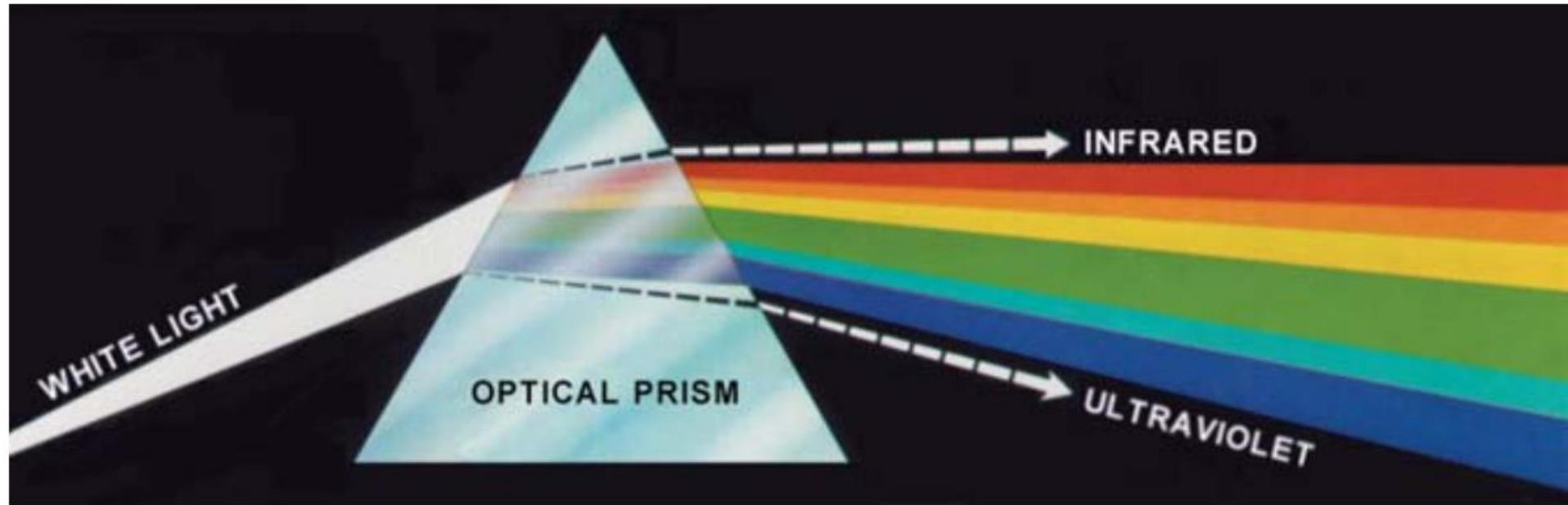


Figure: Digital image (a) Grey scale b) Pseudo color



2.3 Color Fundamentals

In 1666, Sir Isaac Newton discovered that when a beam of sunlight passes through a glass prism, the emerging beam of light is not white but consists instead of a continuous spectrum of colors ranging from violet at one end to red at the other.

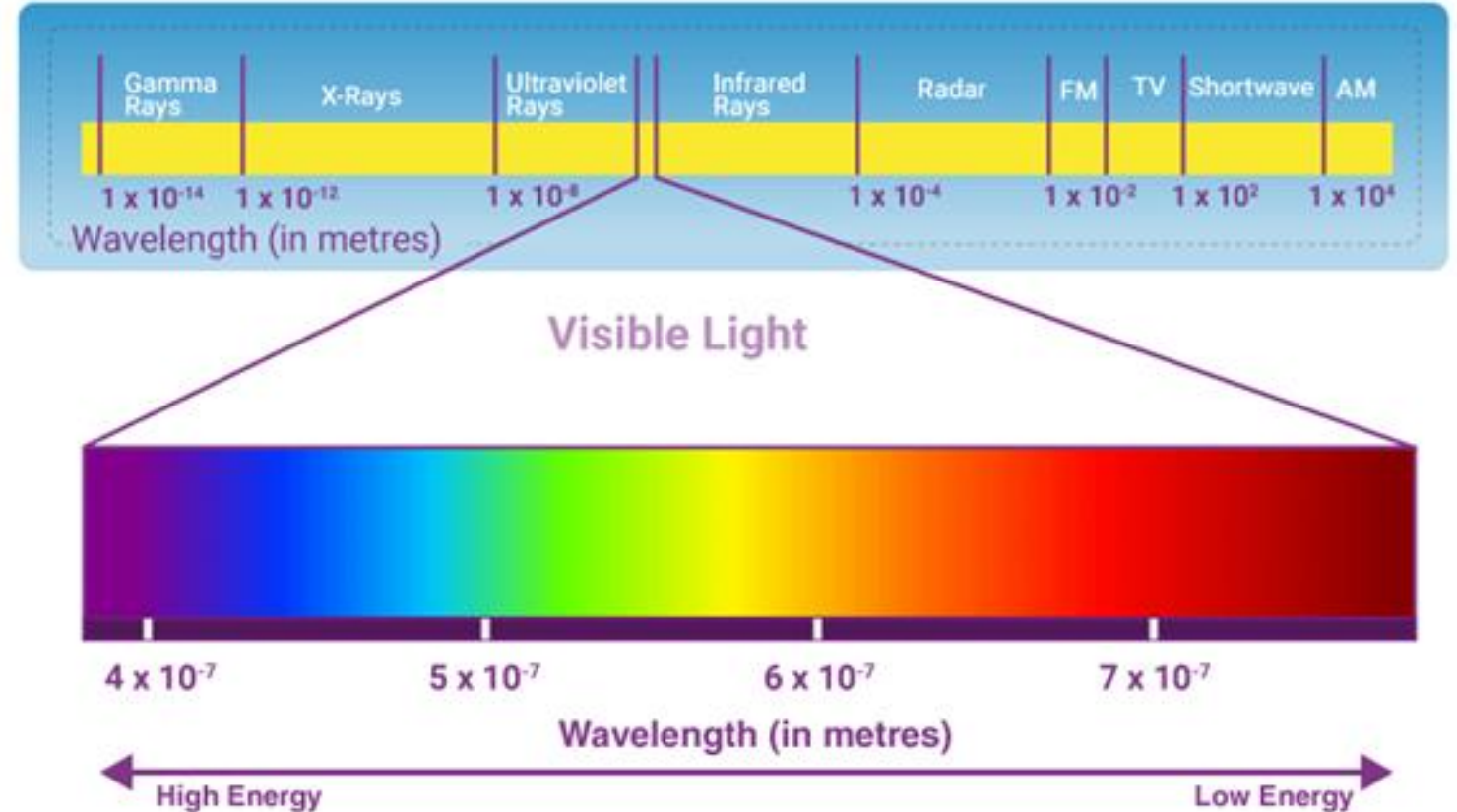


Color spectrum seen by passing white light through a prism.

The color spectrum may be divided into six broad regions: violet, blue, green, yellow, orange, and red.

When viewed in full color, no color in the spectrum ends abruptly, but rather each color blends smoothly into the next.

Basically, the colors that humans and some other animals perceive in an object are determined by the nature of the light reflected from the object.



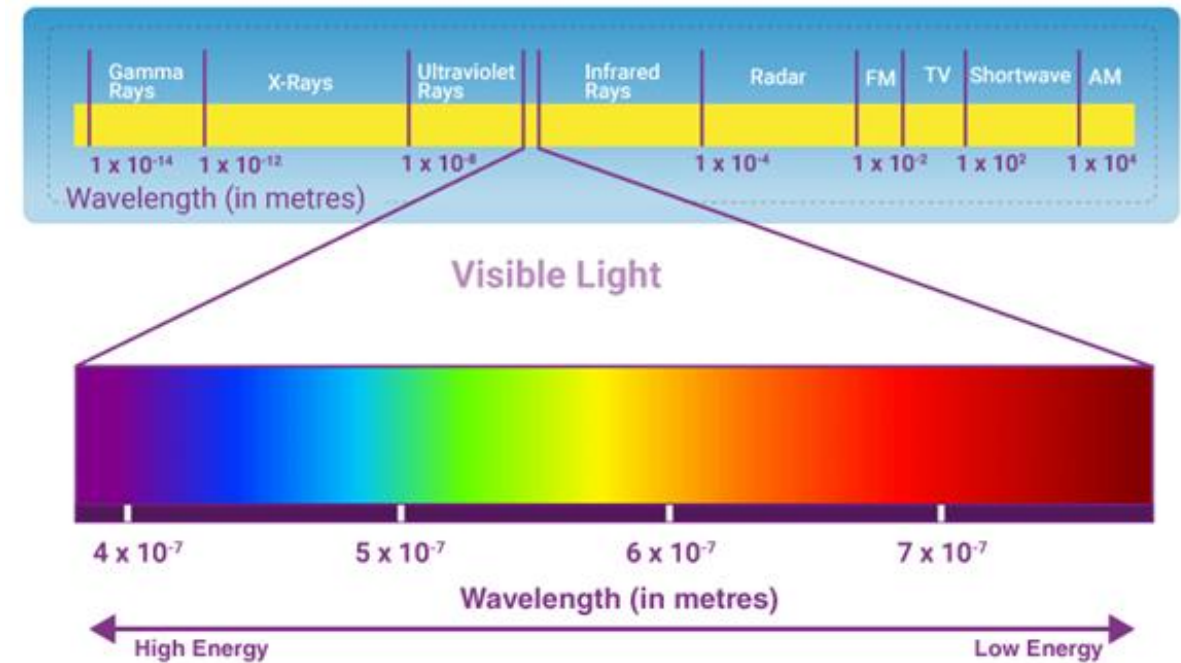
Wavelengths comprising the visible range of the electromagnetic spectrum.

As illustrated in Figure, visible light is composed of a relatively narrow band of frequencies in the electromagnetic spectrum.

A body that reflects light that is balanced in all visible wavelengths appears white to the observer.

However, a body that favors reflectance in a limited range of the visible spectrum exhibits some shades of color.

*** For example, green objects reflect light with wavelengths primarily in the 500 to 570 nm range while absorbing most of the energy at other wavelengths.



2.4 Characterization of light

Characterization of light is central to the science of color. If the light is achromatic (void of color), its only attribute is its intensity, or amount.

2.4.1 Achromatic Light

Achromatic light is what viewers see on a black and white television set, and it has been an implicit component of our discussion of image processing thus far.

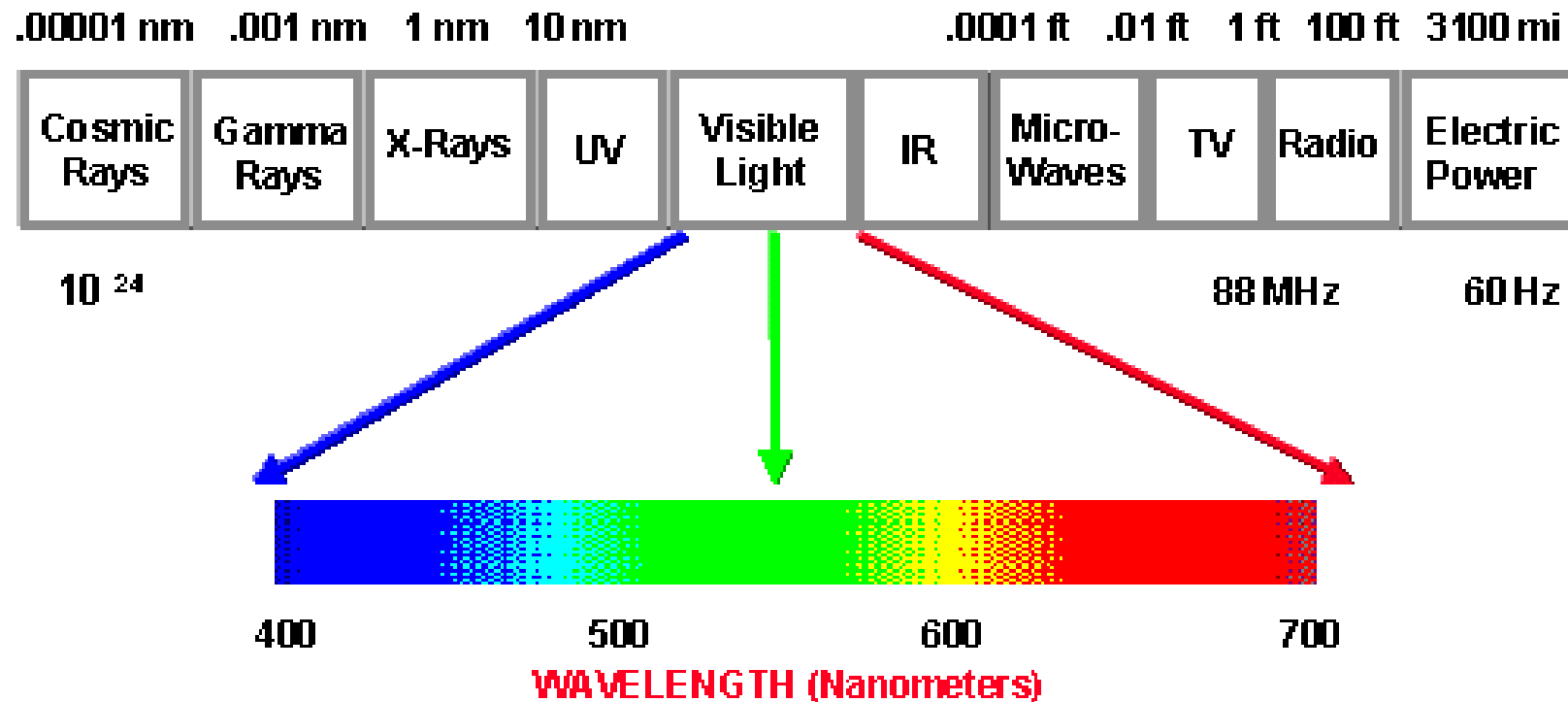
As defined in Chapter 2, and used numerous times since, the term gray level refers to a scalar measure of intensity that ranges from black, to grays, and finally to white.



2.4.2 Chromatic Light

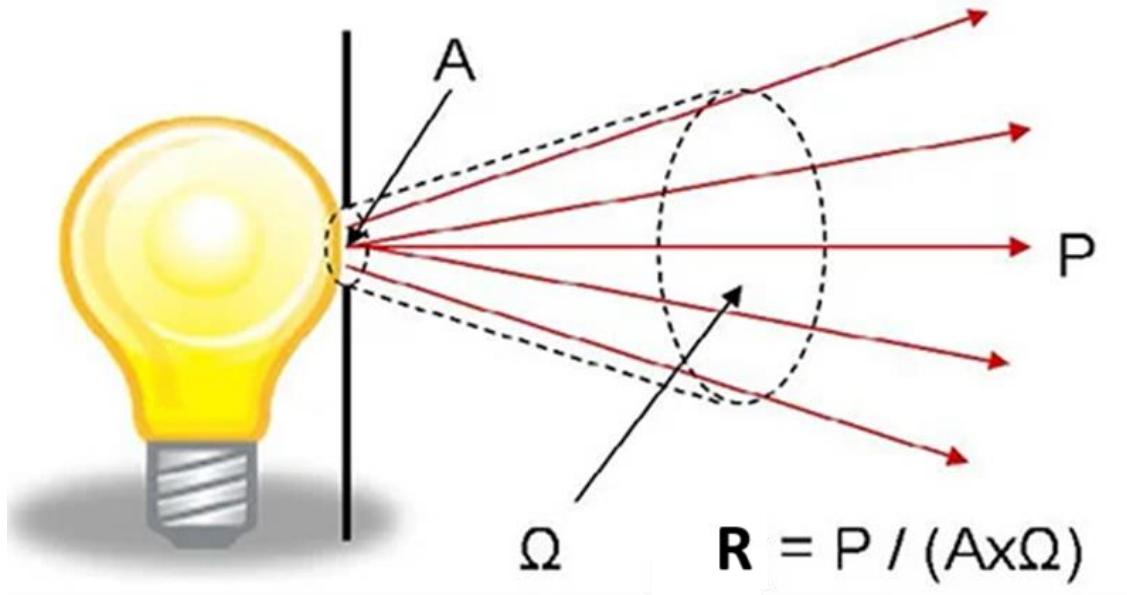
Chromatic light spans the electromagnetic spectrum from approximately 400 to 700 nm.

electromagnetic spectrum



Three basic quantities are used to describe the quality of a chromatic light source: radiance, luminance, and brightness.

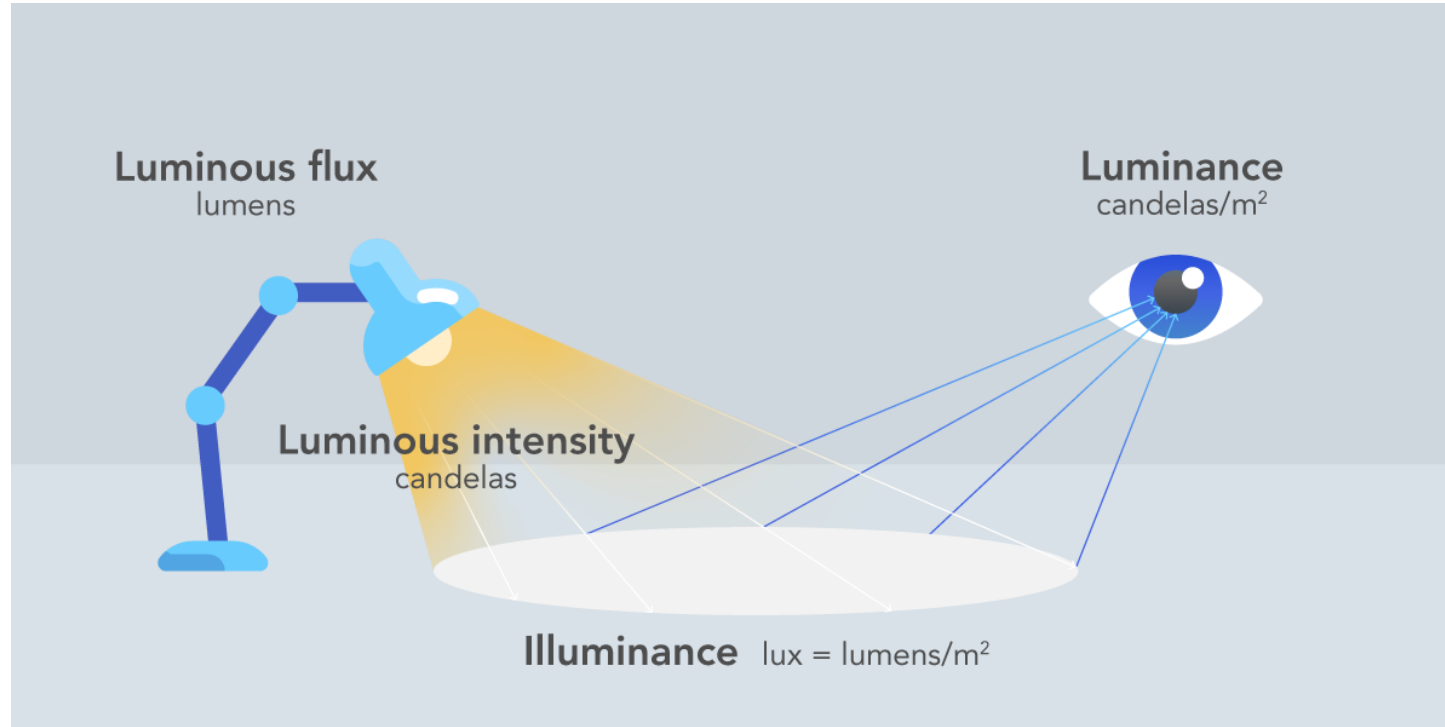
Radiance is the total amount of energy that flows from the light source, and it is usually measured in watts (W).



The radiance (R) of the source emitting area (A) equals the radiation power (P), which is emitted from A and propagates in solid angle Ω , divided by the area A and the solid angle Ω :

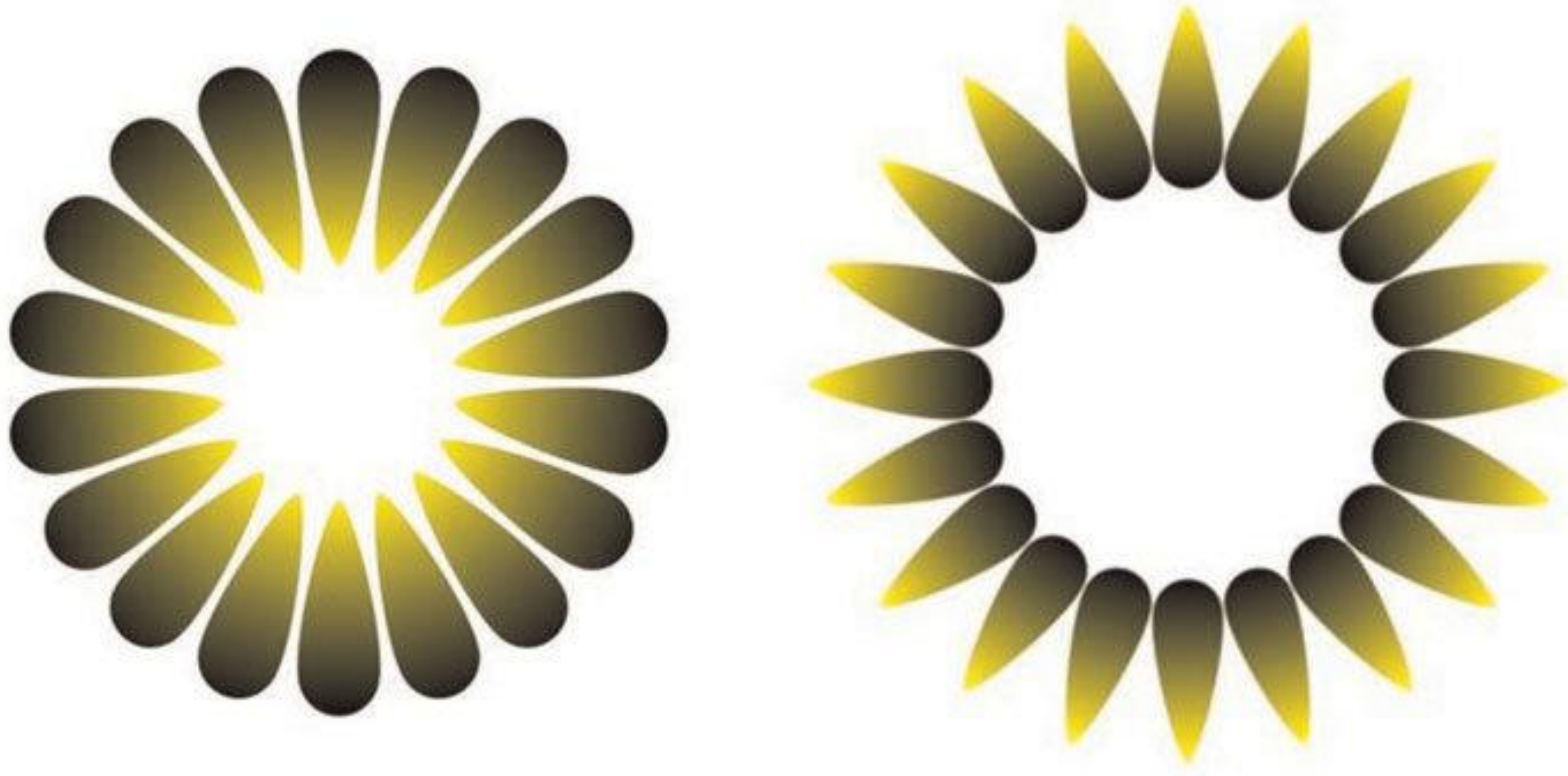
$$R = P / (A \times \Omega).$$

Luminance, measured in lumens (lm), gives a measure of the amount of energy an observer perceives from a light source.



For example, light emitted from a source operating in the far infrared region of the spectrum could have significant energy (radiance), but an observer would hardly perceive it; its luminance would be almost zero.

Brightness is a subjective descriptor that is practically impossible to measure. It embodies the achromatic notion of intensity and is one of the key factors in describing color sensation.



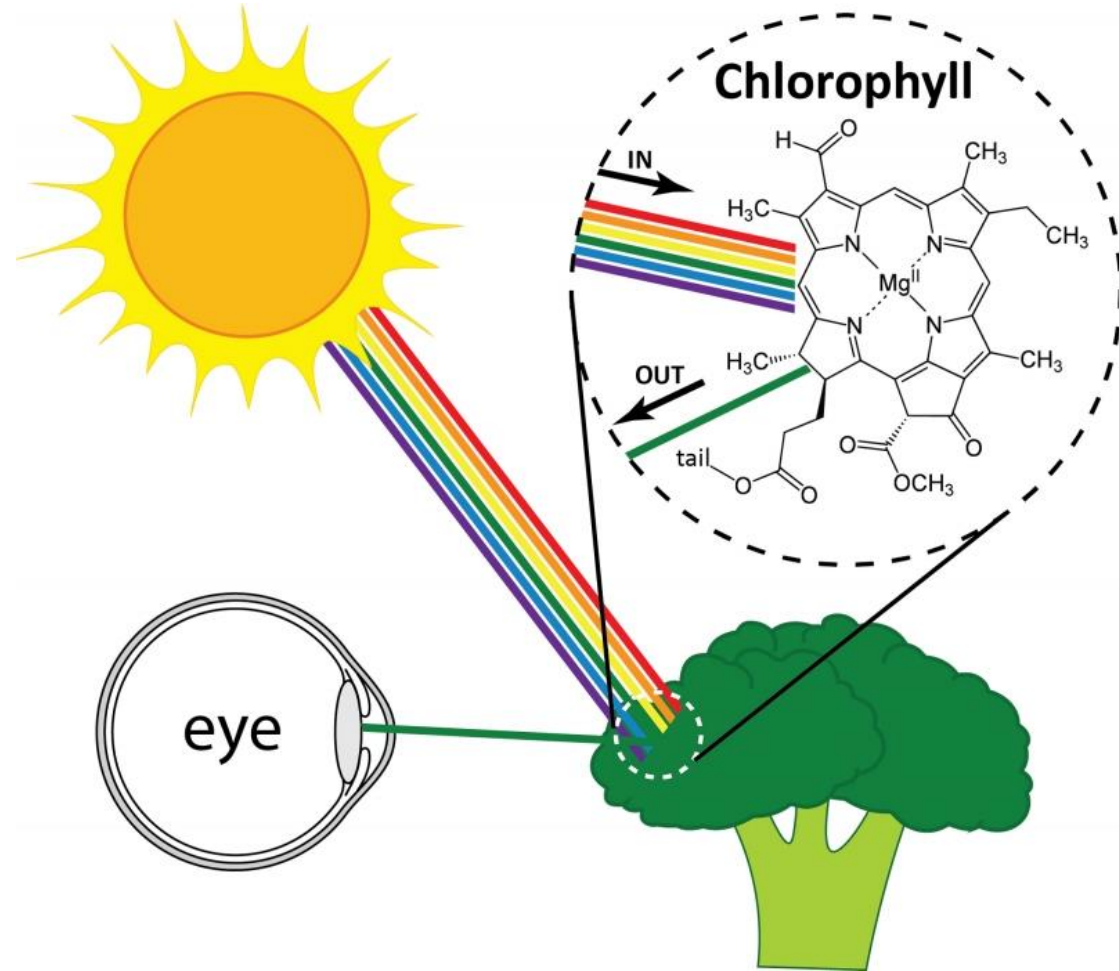
Light and Pigments

When light sources (e.g., red, green, and blue) are combined, they add together to produce new colors.

The more light we add, the brighter the result becomes.

When light hits a pigmented surface (e.g., paint, ink, or dye), some wavelengths are absorbed, and others are reflected.

The reflected light is what we perceive as color.



2.4.2.1 Absorption of light by the red, green, and blue cones in the human eye as a function of wavelength

Cones are the sensors in the eye responsible for color vision.

Detailed experimental evidence has established that the 6 to 7 million cones in the human eye can be divided into three principal sensing categories, corresponding roughly to red, green, and blue.

Approximately 65% of all cones are sensitive to red light, 33% are sensitive to green light, and only about 2% are sensitive to blue.

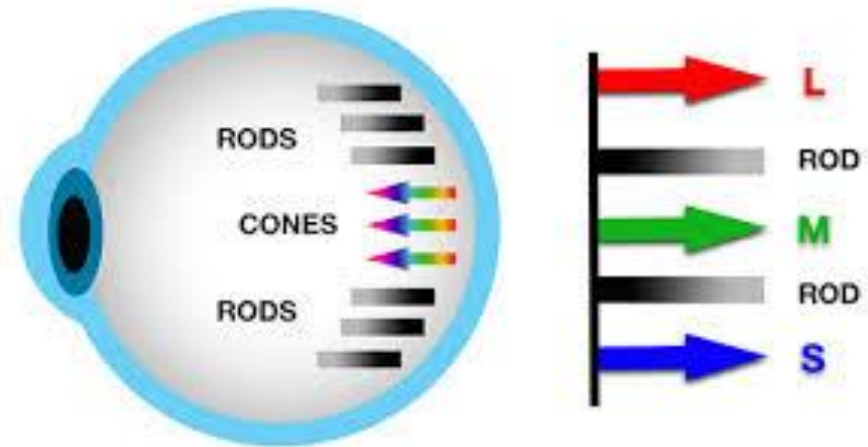
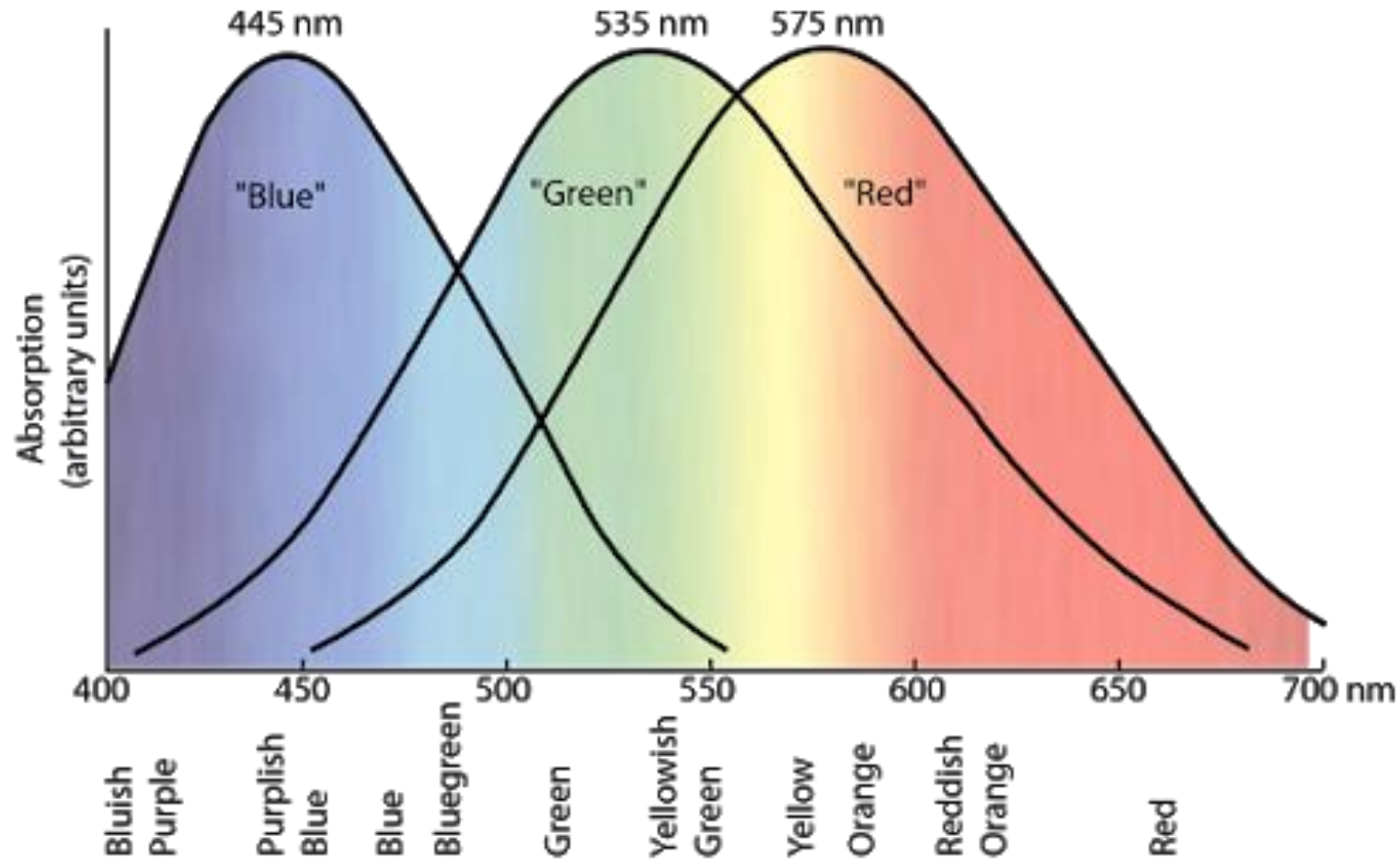


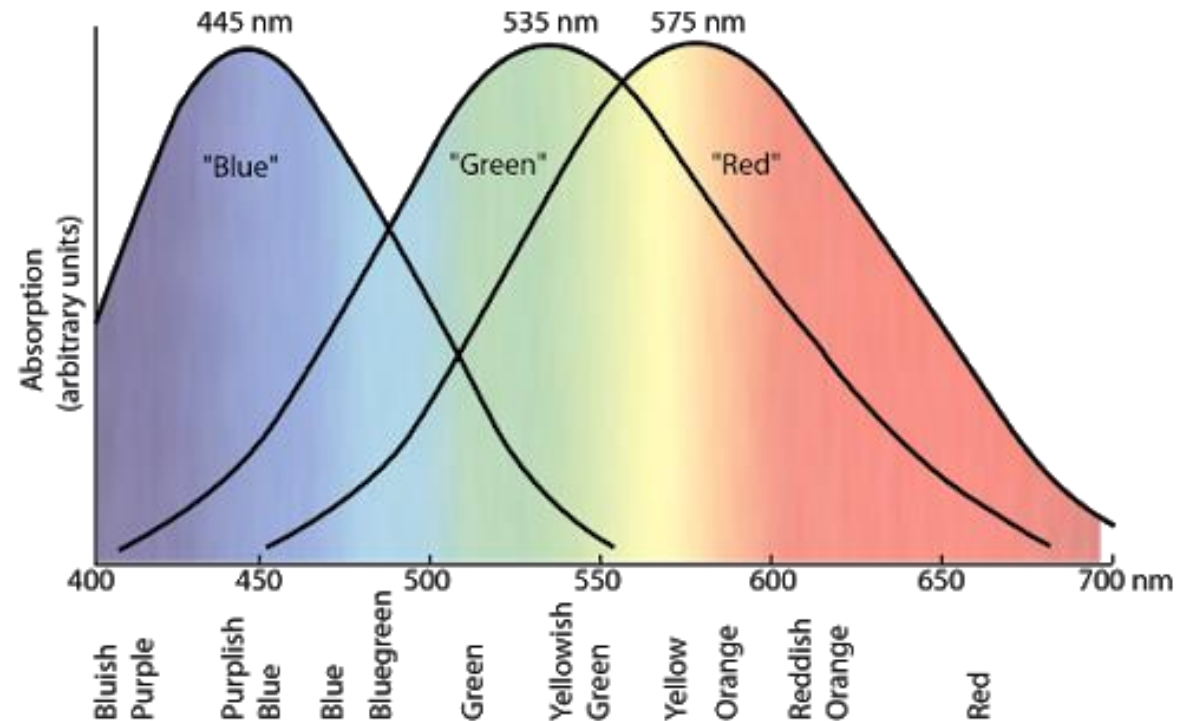
Figure shows average experimental curves detailing the absorption of light by the red, green, and blue cones in the eye.



2.4.2.2 Primary Color

Due to these absorption characteristics of the human eye, colors are seen as variable combinations of the so-called primary colors red (R), green (G), and blue (B).

For the purpose of standardization, the CIE (Commission Internationale de l'Eclairage — the International Commission on Illumination) designated in 1931 the following specific wavelength values to the three primary colors:



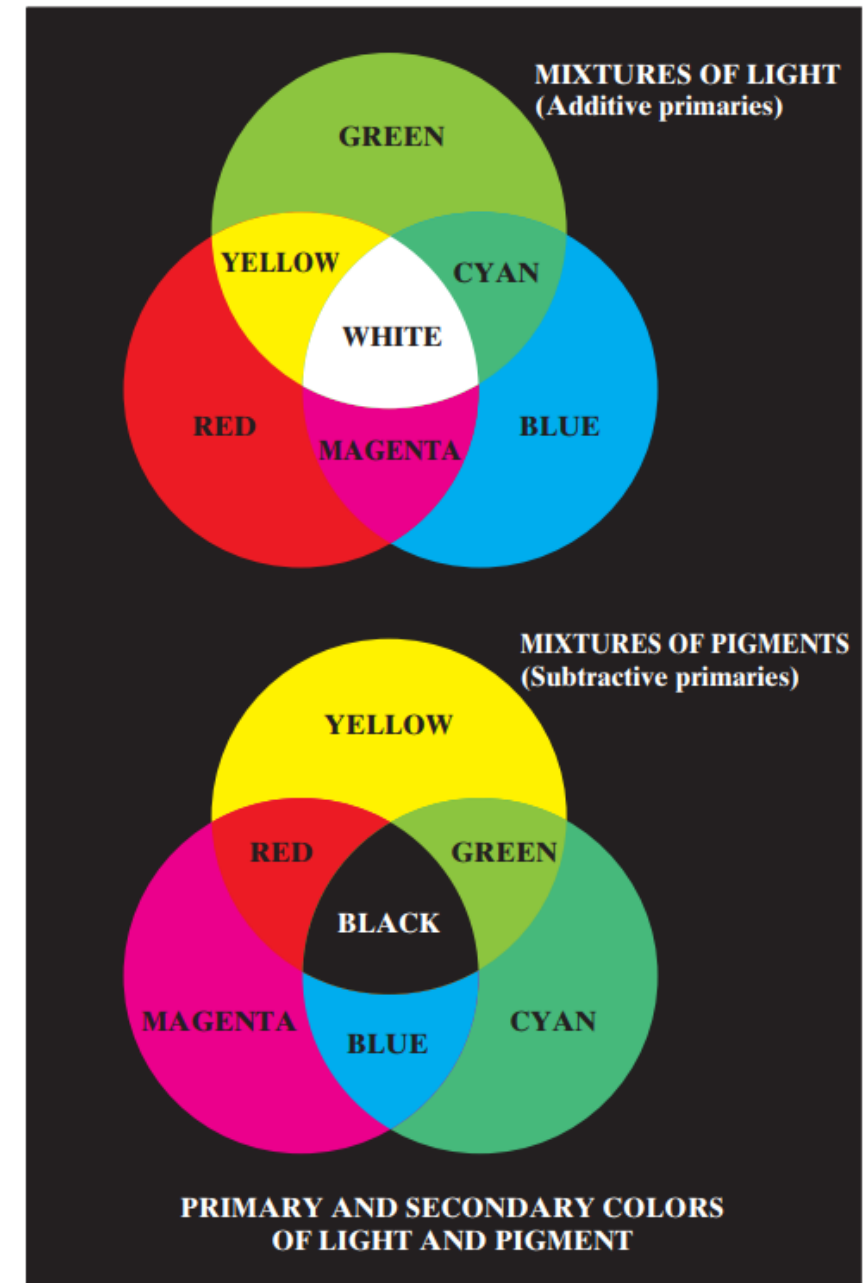
blue = 435.8 nm, green = 546.1 nm and red = 700 nm.

2.4.2.3 Secondary Color from Primary Color

The primary colors can be added to produce the secondary colors of light— magenta (red plus blue), cyan (green plus blue), and yellow (red plus green).

Mixing the three primaries, or a secondary with its opposite primary color, in the right intensities produces white light.

This result is shown in Fig. which also illustrates the three primary colors and their combinations to produce the secondary colors.



2.4.2.4 Distinguish color from one another

The characteristics generally used to distinguish one color from another are brightness, hue, and saturation.

Brightness embodies the achromatic notion of intensity.

Hue is an attribute associated with the dominant wavelength in a mixture of light waves. Hue represents dominant color as perceived by an observer.

Thus, when we call an object red, orange, or yellow, we are referring to its hue.



Saturation refers to the relative purity or the amount of white light mixed with a hue.

The pure spectrum colors are fully saturated.

Colors such as pink (red and white) and lavender (violet and white) are less saturated, with the degree of saturation being inversely proportional to the amount of white light added.



Hue and saturation taken together are called **chromaticity**, and, therefore, a color may be characterized by its brightness and chromaticity.

The amounts of red, green, and blue needed to form any particular color are called the **tristimulus values** and are denoted, X, Y, and Z, respectively.

A color is then specified by its trichromatic coefficients, defined as

$$x = \frac{X}{X + Y + Z}$$

$$y = \frac{Y}{X + Y + Z}$$

$$z = \frac{Z}{X + Y + Z}$$

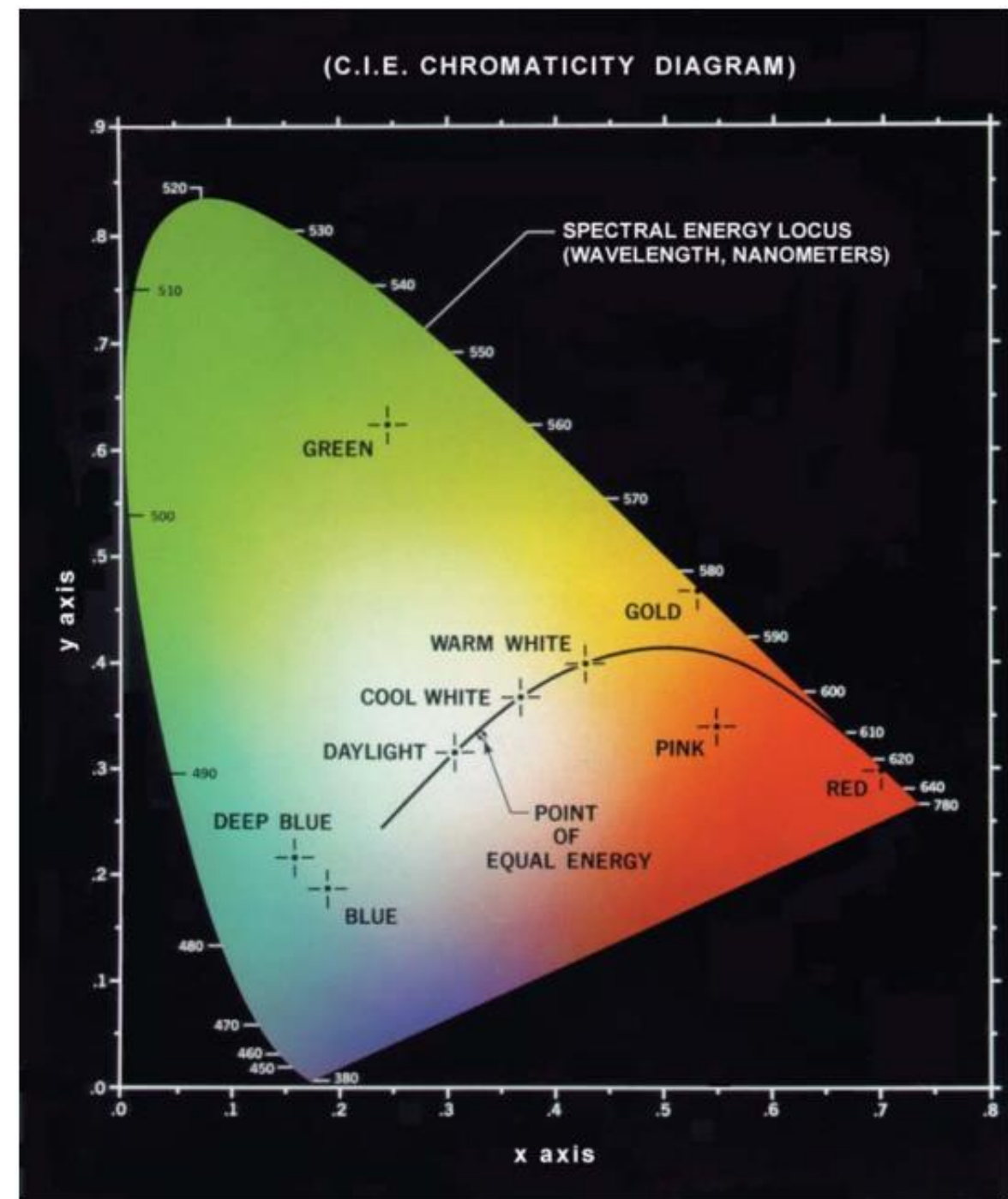
$$x + y + z = 1$$

For any wavelength of light in the visible spectrum, the tristimulus values needed to produce the color corresponding to that wavelength can be obtained directly from curves or tables that have been compiled from extensive experimental results.

Another approach for specifying colors is to use the **CIE chromaticity diagram** which shows color composition as a function of x (red) and y (green).

For any value of x and y, the corresponding value of z (blue) is obtained from Eqn:

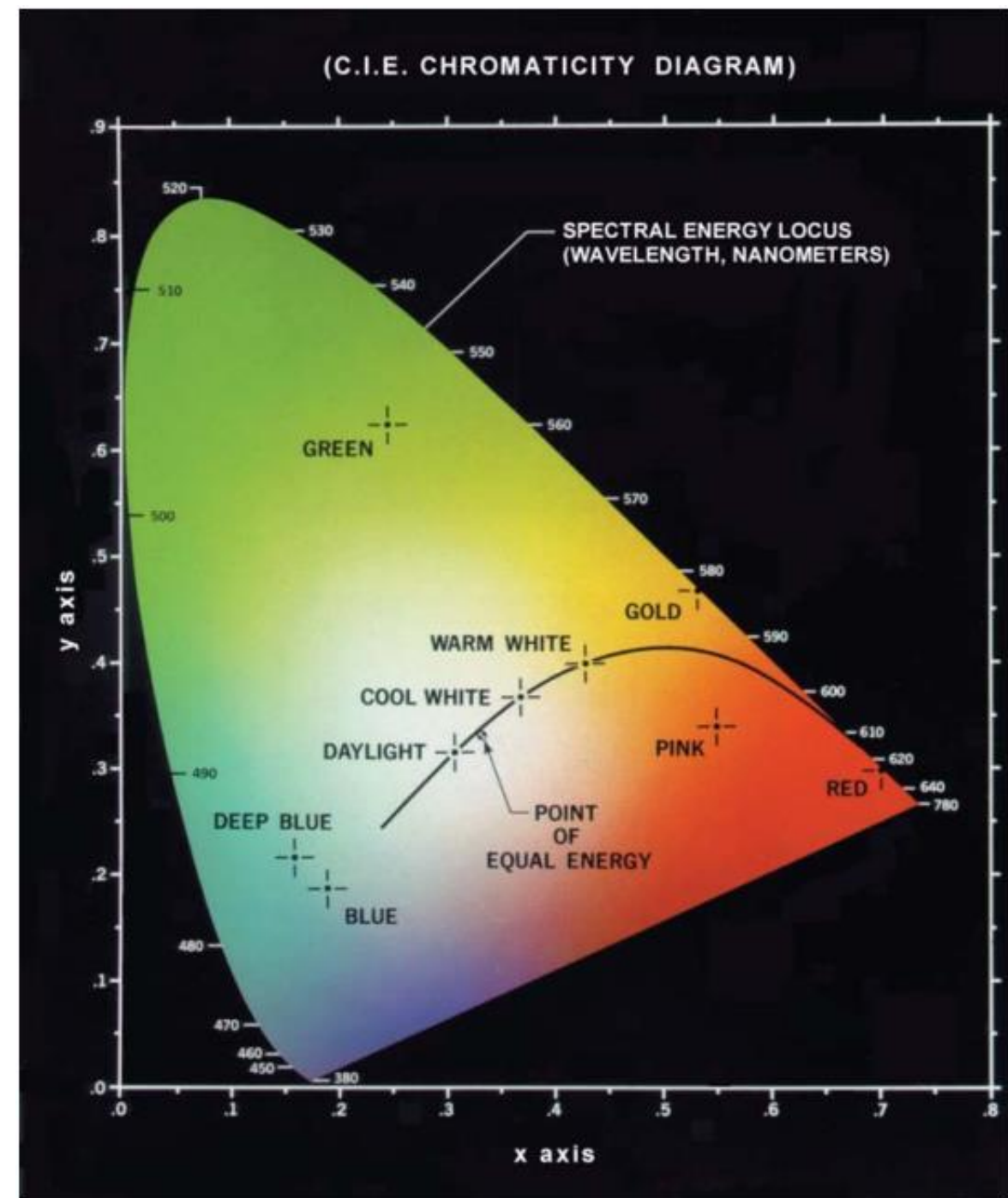
$$z = 1 - (x + y)$$



The *point marked green* in Fig., for example, has approximately 62% green and 25% red content.

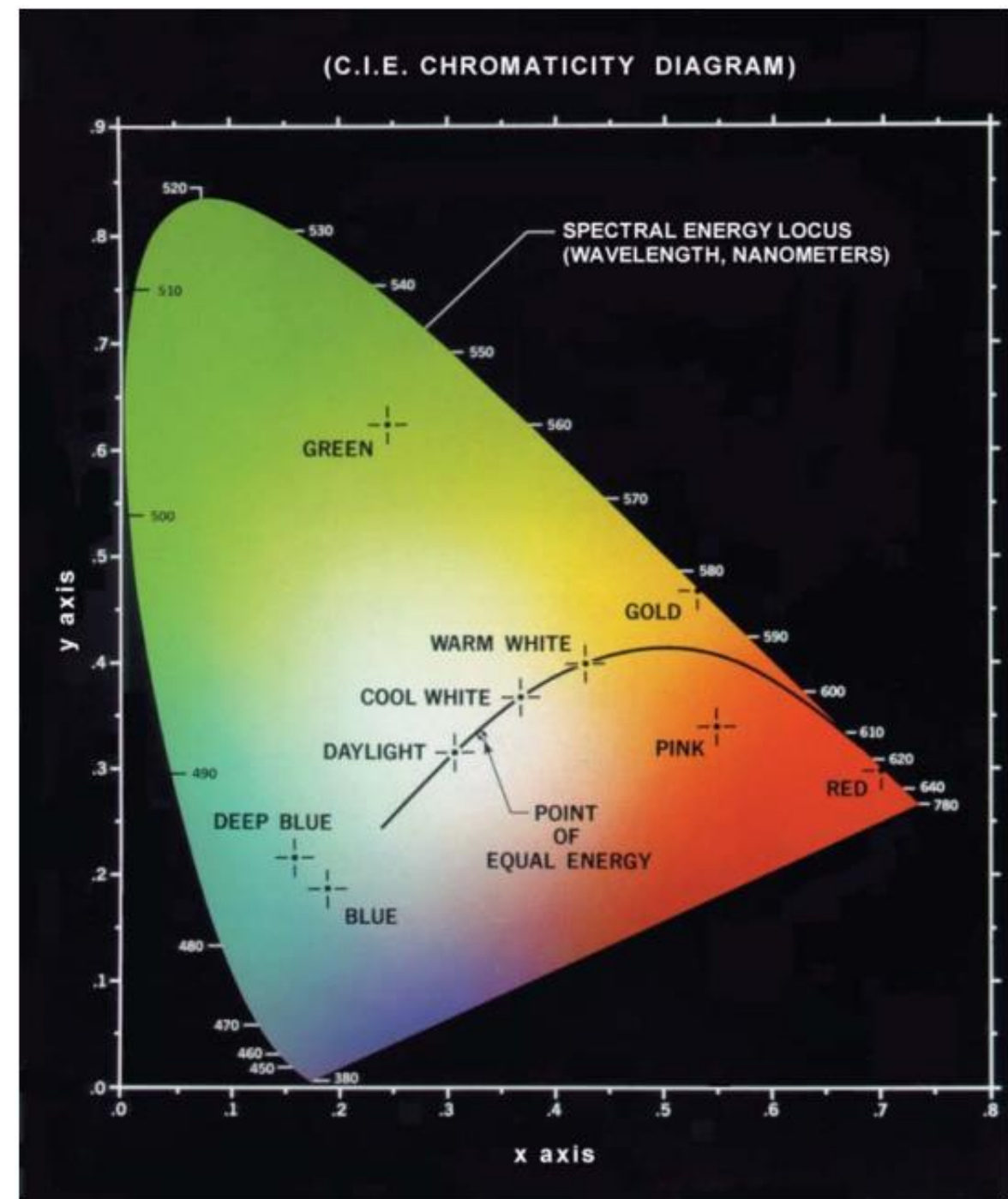
$$z = 1 - (x + y)$$

From Eq., the composition of blue is approximately 13%

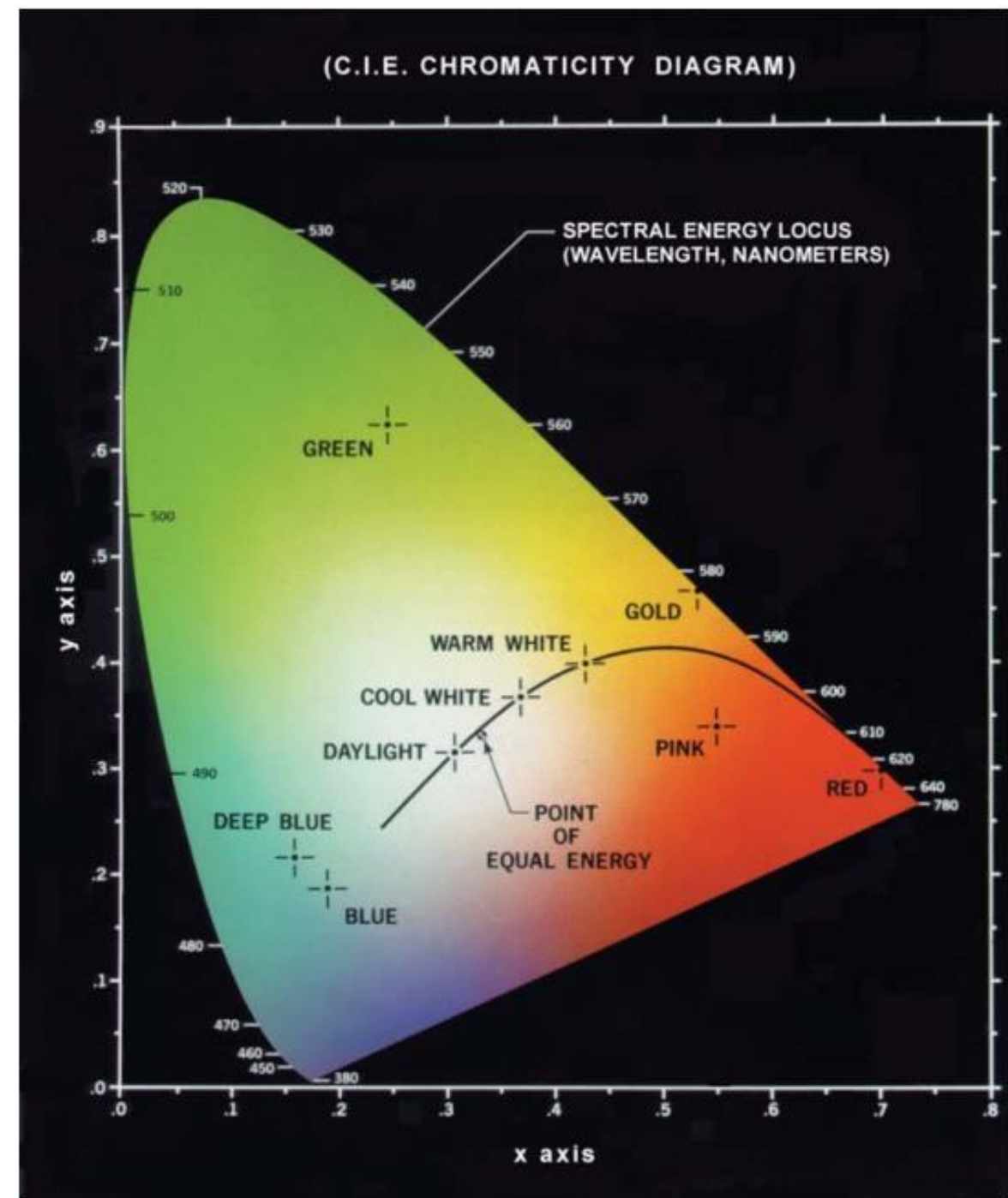


The positions of the various spectrum colors—from violet at 380 nm to red at 780 nm—are indicated around the boundary of the tongue-shaped chromaticity diagram.

*** Any point not actually on the boundary but within the diagram represents some mixture of spectrum colors.



- ✓ Any point located on the boundary of the chromaticity chart is fully saturated.
- As a point leaves the boundary and approaches the point of equal energy, more white light is added to the color and it becomes less saturated.
- ❖ The saturation at the point of equal energy is zero.



The chromaticity diagram is useful for color mixing because a straight-line segment joining any two points in the diagram defines all the different color variations that can be obtained by combining these two colors additively.

Consider, for example, a straight line drawn from the red to the green points shown in Fig. If there is more red light than green light, the exact point representing the new color will be on the line segment, but it will be closer to the red point than to the green point.

